

# A RANDOM WALK APPROACH FOR DERIVING EXACT EXPRESSIONS FOR PROBABILITIES ASSOCIATED WITH EFRON'S BIASED COIN DESIGN

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**Summary:** Markaryan and Rosenberger (2010) derived exact expressions for probabilities associated with Efron's biased coin design by induction, using subtle but somewhat lengthy combinatorial arguments. Concise derivations of these expressions are presented here and are rooted in the geometry of one-dimensional random walks.

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## 1. Introduction

Suppose that patients in a clinical trial are to be allocated sequentially to one of two treatments,  $A$  or  $B$ . An immediate problem in the planning of such a trial is to devise an allocation scheme which maintains a balance between the numbers of patients allocated to the two treatments throughout the trial and at the same time controls experimental bias. Efron addressed this problem in a seminal paper in 1971 by proposing an allocation procedure which he termed the biased coin design (BCD). Specifically, the scheme proceeds as follows: if  $N_A$  patients have been assigned to treatment  $A$  and  $N_B$  to treatment  $B$  at a particular time point in the trial, then the probability of allocating the next patient to treatment  $A$  is given by  $p$  if  $N_B > N_A$ , by  $\frac{1}{2}$  if  $N_B = N_A$  and by  $q$  if  $N_B < N_A$  where  $0 \leq p \leq 1$  and  $q = 1 - p$ . Efron (1971) recommended a working probability of  $p = \frac{2}{3}$ .

The BCD has received much attention in the literature and its asymptotic properties are well understood. In practice however the number of patients in a clinical trial is usually quite small and the elucidation of the exact properties of the design is therefore of key importance. Markaryan and Rosenberger (2010) recently reported the derivation of such properties, with exact expressions for the probability that, after  $n = N_A + N_B$  patients have entered the trial, the difference in the numbers of patients allocated to the two treatments,  $D_n = N_A - N_B$ , is equal to  $k$ , where  $-n \leq k \leq n$ , at the core of their arguments. They developed these expressions by induction, using subtle but somewhat lengthy combinatorial arguments. In the present note it is shown that the requisite expressions can be derived succinctly by adopting an approach gleaned from the classical theory of one-dimensional random walks.

## 2. Derivations

Consider the formulae (2.1) and (2.2) from Theorem 2.1 of Markaryan and Rosenberger (2010) which give the exact expressions for the probabilities  $P(D_n = k)$  for  $0 < k \leq n$  and  $k = 0$  respectively and, by symmetry, for  $-n \leq k < 0$ . These expressions can be reformulated within the context of a classical one-dimensional random walk by setting  $N_A = a$  and  $N_B = b$  and taking  $n = a + b$  and  $k = a - b \geq 0$  where  $a$  and  $b$  are the numbers of successes (“up”) and failures (“down”) respectively. Then formula (2.1) can be re-expressed as

$$P(D_{a+b} = a - b) = \frac{1}{2} p^b \sum_{j=0}^b \frac{a-j}{a+j} \binom{a+j}{j} q^{a-b+j-1} \text{ for } a > b \quad (1)$$

and formula (2.2) as

$$P(D_{2a} = 0) = p^a \sum_{j=0}^{a-1} \frac{a-j}{a+j} \binom{a+j}{j} q^j, \quad (2)$$

since in the latter case  $a = b$ .

The derivation of formula (2) follows immediately from Theorem 4, Chapter III, page 90 of Feller (1968). Specifically, suppose that a representative path of length  $n = 2a$  from the origin to the point  $(2a, 0)$  has all sides above the  $t$ -axis and  $r = a - j$  points, including the endpoint  $(2a, 0)$ , on that axis. Then the total number of paths generated by all such representative paths is given by

$$\frac{r}{n-r} \binom{n-r}{\frac{n}{2}} 2^r = \frac{a-j}{a+j} \binom{a+j}{a} 2^{a-j}.$$

Furthermore, a little reflection shows that, in the context of Efron’s BCD, each path is associated with a probability of

$$\left(\frac{1}{2}\right)^{a-j} p^a q^j.$$

Thus, since  $j$  can range from 0 to  $a - 1$ , it follows that

$$P(D_{2a} = 0) = p^a \sum_{j=0}^{a-1} \frac{a-j}{a+j} \binom{a+j}{j} q^j$$

as required.

The derivation of formula (1) is a little more intricate than that of formula (2) but nevertheless follows the same structure, namely that of Theorem 4, Chapter III, page 90 of Feller (1968). Specifically, consider a representative path from the origin to the point  $(a+b, a-b)$ , where  $a > b$ , with all sides above the  $t$ -axis and with exactly  $b-j$  points lying on that axis,  $j = 0, 1, \dots, b$ . Each section of this representative path between endpoints, that is points lying on the axis, can be reflected about the  $t$ -axis so that the path represents a total of  $2^{b-j}$  paths. Consider now creating a new path by deleting all sides of the representative path which have their rightmost point on the  $t$ -axis. The resultant path is of length  $(a+b) - (b-j) = a+j$ , has accumulated gain  $(a-b) - (b-j) = a-j > 0$  and lies strictly above the  $t$ -axis. Thus, by the Ballot Theorem, as stated in Chapter III, page 73 of Feller

(1968), there are  $\frac{a-j}{a+j} \binom{a+j}{j}$  such representative paths, thereby generating a total of

$$\frac{a-j}{a+j} \binom{a+j}{j} \times 2^{b-j}$$

paths from the origin to the point  $(a+b, a-b)$  with  $a > b$  and with exactly  $b-j$  points on the  $t$ -axis. Observe also that, within the context of the BCD, the probability associated with each of these paths is given by

$$\frac{1}{2} \left( \frac{1}{2} \right)^{b-j} p^b q^{a-(b-j)-1}.$$

Thus, since  $j$  ranges from 0 to  $b$ , it now follows that

$$P(D_{a+b} = a-b) = \frac{1}{2} p^b \sum_{j=0}^b \frac{a-j}{a+j} \binom{a+j}{j} q^{a-b+j-1}$$

as required.

### 3. Conclusions

Concise derivations of exact expressions for probabilities associated with Efron's BCD are presented here and are rooted in the arguably "old-fashioned" geometry of one-dimensional random walks (Feller, 1968). It is tempting to suggest that the approach can also be used to derive exact expressions for probabilities relating to, for example, generalized BCDs and response-adaptive urn models. However in such cases the allocation of a treatment to a patient during the trial depends on the numbers of patients already assigned to the two treatments and not on their difference. Thus the connection with one-dimensional random walks would seem, at least at first glance, to be lost.

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